MILITARY SPECIFICATION

# CAP, NOSE
# FOR
# ROCKET, HE, 66MM, ANTI-TANK, M72
# ASSEMBLING AND PACKING

## 1. SCOPE

1.1 This specification covers special requirements not normally required by the drawings and provides the quality assurance provisions for the fabrication of parts, assembling and packing for one type of piezoid electrical power generating nose cap assembly for use in the Rocket, HE, 66MM, Anti-Tank, M72.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

| | |
|---|---|
| MIL-A-2550 | - Ammunition and Special Weapons; General Specification for. |
| MIL-D-45608 | - Designs and Drawings, Ordnance Inspection Equipment. |
| MIL-I-45607 | - Inspection Equipment, Supply and Maintenance of. |
| MIL-P-10025 | - Packaging, Packing and Marking for Interplant Shipment of Inert Ammunition Components; General Specification for. |
| MIL-P-12815 | - Power Source, Piezoid (Lucky). |

FSC: 1340

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Quality Assurance Terms and Definitions.
MIL-STD-1167 - Ammunition Data Cards.
MIL-STD-1168 - Lot Numbering of Ammunition.
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ORDNANCE CORPS

10048528 - Nose Cap Assembly.

PUBLICATIONS

ORDNANCE CORPS

IEL-10048528 - Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Assembly.-The nose cap assembly shall comply with all requirements specified on Drawing (dwg.) 10048528, and with all requirements specified in applicable specifications.

3.3 Electrical output.-The nose cap assembly, when impacted on the closed end by a 1.94 ounce steel ball dropped from a height of 2 inches maximum, shall produce electrical energy equivalent to the energy discharged by a 0.005 microfarad capacitor which had been charged to 9 volts, minimum.



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3.4 Workmanship.-All parts shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.
b. Number of units of product inspected.
c. Results obtained, by defect code, for all inspections performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.
f. Number of items in the lot.
g. Date submitted.



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The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 Government verification.-Using the contractor's written quality assurance procedure, this detailed specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Initial production

4.2.1 Submission.-At the start of regular production, the contractor shall submit an initial production sample to a Government approved facility as designated by the contracting officer (see 6.4) for evaluation in accordance with the provisions of 4.2.2. The initial production sample shall consist of:

a. 25 complete nose cap assemblies
b. 25 sets of each subassembly described by the applicable drawings.
c. 25 complete sets of parts.

The initial production sample shall consist of material that has been produced by the contractor or furnished by a supplier and manufactured using the same production process, procedures, and equipment that will be used in fulfilling the contract. All materials shall be obtained from the same sources of supply that will be used in regular production. The initial production sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply so as to significantly affect product uniformity as determined by the Government. Prior to submission, the



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contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects shall not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspections to be performed.-The entire sample shall be examined and inspected by the Government for every dimension and requirement on the applicable drawings. The sample shall then be inspected for those characteristics listed in 4.3.2 and 4.3.3.1 which are in addition to the drawing requirements. The complete nose cap assemblies shall be subjected to the testing specified in 4.3.3.2, 4.3.3.3 and 4.3.3.4 without regard to sample size or AQL.

4.2.3 Rejection.-If any portion of the initial production sample fails to conform to any of the applicable requirements, the initial production sample shall be rejected. The Government reserves the right to terminate its inspection upon observation of any failure of any portion of the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new initial production sample or portion thereof. The contractor is in no way authorized by the Government to initiate regular production until a completely acceptable initial production sample is submitted and approved, unless otherwise directed by the contracting officer.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, specification, or revision thereof. Changes to either the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government inspector, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL-STD-105 and shall be numbered in accordance with MIL-STD-1168.



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Unless otherwise approved by the contracting officer, the inspection lot size of major assemblies or end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest quantity contractually scheduled for delivery during any month of the contract period. Inspection lots for components or subassemblies, other than the items of delivery, shall be homogeneous and of a size convenient to the contractor.

4.3.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Standard MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. Once agreement has been reached as to which method will be used (group or individual) that method will be used for all of the CD's listed in this specification. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Resistor (see dwg. 10048645 covering a detail of dwg. 10048671).

| Categories | Defects | Method of Inspection | Code No. (see 6.2) |
|---|---|---|---|
| Critical: None defined. | | | |
| Major: | AQL 0.40 percent | | |
| 101. | Resistance | Test | 01001 |
| Minor: | AQL 0.65 percent | | |
| 201. | Evidence of poor workmanship (see 3.4) | Visual | 01002 |

4.3.2.2 Pin, terminal, ground (see dwg. 10048706 covering a detail of dwg. 10048646).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: None defined. | | | |



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| | | | |
|---|---|---|---|
| Major: | AQL 0.40 percent | | |
| 101. | Large outside diameter..... | Gage | 02001 |
| 102. | Plating missing or incomplete.. | Visual | 02002 |
| Minor: | AQL 1.00 percent | | |
| 201. | Total length, minimum (min.)... | Gage | 02003 |
| 202. | Evidence of poor workmanship (see 3.4)..... | Visual | 02004 |

4.3.2.3 Terminal, positive (see dwg. 10048726 covering a detail of dwg. 10048528).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | AQL 0.40 percent | | |
| 101. | Outside diameter to flange..... | Gage | 03001 |
| 102. | Plating missing or incomplete.. | Visual | 03002 |
| Minor: | AQL 1.00 percent | | |
| 201. | Total length, min..... | Gage | 03003 |
| 202. | Evidence of poor workmanship (see 3.4)..... | Visual | 03004 |

4.3.2.4 Disc, ceramic (see dwg. 10048640 covering a detail of dwg. 10048671).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | AQL 0.40 percent | | |
| 101. | Diameter of silver coating over maximum (max.)..... | Gage | 04001 |
| 102. | Silver coating not clear of edge of ground terminal slot and centerhole..... | Visual | 04002 |
| Minor: | AQL 0.65 percent | | |
| 201. | Evidence of poor workmanship (see 3.4)..... | Visual | 04003 |



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4.3.2.5 Flat lucky and terminal assembly (see dwg. 10048670 covering a detail of dwg. 10048528).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | AQL 0.40 percent | | |
| 101. | Conductive cement on unsilvered portion of lucky | Visual | 05001 |
| Minor: | AQL 0.65 percent | | |
| 201. | Evidence of poor workmanship (see 3.4) | Visual | 05002 |

4.3.2.6 Nose cap (see dwg. 10048669 covering a detail of dwg. 10048528).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | AQL 0.65 percent | | |
| 101. | Inside diameter of open end, max | Gage | 06001 |
| 102. | Length to tip of locking tabs | Gage | 06002 |
| 103. | Height of locking tab from inside cylindrical surface | Gage | 06003 |
| 104 | Potting detents missing | Visual | 06004 |
| Minor: | AQL 1.50 percent | | |
| 201. | Inside diameter of open end, min | Gage | 06005 |
| 202. | Angular location of locking tabs | Gage | 06006 |
| 203. | Evidence of poor workmanship (see 3.4) | Visual | 06007 |



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4.3.2.7 Assembly, nose cap (see dwg. 10048528).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: None defined. | | | |
| Major: | AQL 0.65 percent | | |
| 101. | Concentricity of positive terminal with inside diameter of open end | Gage | 07001 |
| 102. | Location of ground terminal with respect to the positive terminal and locking tab | Gage | 07002 |
| 103. | Plating damaged | Visual | 07003 |
| 104. | Terminals not free of foreign material | Visual | 07004 |
| Minor: | AQL 1.50 percent | | |
| 201. | Length to terminals | Gage | 07005 |
| 202. | Length to potting compound, min | Gage | 07006 |
| 203. | Paint incomplete, missing, or damaged | Visual | 07007 |
| 204. | Evidence of poor workmanship (see 3.4) | Visual | 07008 |

4.3.3 Testing

4.3.3.1 Test of lucky, less resistor (see dwg. 10048640) - Major defect - Code No. 04004.-Sampling, testing, and acceptance criteria shall be as specified in Specification MIL-P-12815.

4.3.3.2 Terminal resistance (see dwg. 10048528) - Major defect - Code No. 07009.-This test shall be performed 100 percent. Any nose cap assemblies which are classed defective shall be removed from the lot. This test shall be performed as specified in 4.4.1 using test equipment specified in 4.3.4.

4.3.3.3 Electrical output (see 3.3) - Major defect - Code No. 07010.-This test shall be performed 100 percent. The test shall be performed as specified in 4.4.2 using test equipment specified in 4.3.4. Any nose cap assembly which fails to comply with the specified requirement shall be classed defective and removed from the lot.



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4.3.3.4 Visual examination for marking on tab locating the ground terminal pin (see dwg. 10048528) - Major defect - Code No. 07011.-This inspection shall be performed 100 percent. Any nose cap assembly without marking as specified shall be classed defective. Defective nose cap assemblies shall be removed from the lot.

4.3.4 Inspection Equipment.-Index of Inspection Equipment Number IEL-10048528 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL-I-45607.

4.3.4.3 Government use of contractor's inspection and test equipment.-The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test methods and procedures.

4.4.1 Terminal resistance.-The resistance of the nose cap assembly shall be measured between the positive terminal and the ground pin. The assembly shall be classed defective if the electrical resistance fails to comply with the specified requirement.



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4.4.2 Electrical output.-The assembled nose cap assembly shall be attached to the test equipment and a 1/16 inch thick steel disc, 3/4 inch in diameter, shall be placed on the closed, flat end of the assembly which must be level and facing up. The steel ball shall be dropped on the steel disc and the test equipment indication observed. The assembly shall be classed defective if it fails to comply with the test equipment requirements.

## 5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.-The nose cap assemblies shall be packaged, packed and marked in accordance with Specification MIL-P-10025.

## 6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Ammunition data cards shall be prepared for each lot in accordance with MIL-STD-1167.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

6.3.1 Inspection equipment lists.-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (OO Form 1242-3) the numbers of drawings of existing equipment designs or codes as indicated in paragraph 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the



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equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5, and 6.3.6 will be designated in the IEL by the following codes:

CDAF - Contractor's design responsibility on Army format in accordance with MIL-D-45608.
CDCF - Contractor's design responsibility on contractor format.
AD - Army design.
ADMU - Army design, mandatory for use.
CE - Commercial equipment.
SCD - Specification control drawing.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications, mandatory and non-mandatory. When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs (see 6.3.6) or commercial equipment selections (see 6.3.5) must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 Specification control drawing.-Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the



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requirements of specification control drawings must be described in sufficient detail to permit identification and evaluation by the Government (see 6.3.5). Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 Commercial equipment. - Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors normally are not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. When commercial inspection equipment requires the use of special fixturing, the design of this fixturing shall be considered as part of the total design, and 6.3.6 shall apply. In the latter case, only those portions of the commercial equipment need be detailed in the designs as are necessary to completely fabricate, calibrate, and operate the total item of inspection equipment. Lists and fixturing designs shall be submitted for approval as specified in 6.3.7.

6.3.6 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDAF shall comply with the format and requirements of MIL-D-45608, and in addition, contain the detail and information specified above. Final approved contractor designs identified as CDCF shall be suitable for microfilming. They must be non-reproducible diazo or electrostatic prints of top quality full size with black lines on white background and shall provide a legible print produced via the 35MM microfilm process. Contractors shall submit one copy of the final design as a rolled set or flat set. Designs shall be submitted for approval as specified in 6.3.7.



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6.3.7 Submission of contractor designs.-Unless otherwise specified on the Inspection Equipment List, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the Contracting Officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Submission of initial production samples.- Instructions as to the location for the evaluation of the initial production sample shall be obtained from the contracting officer. Upon receipt of such request, the contracting officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquiries should be forwarded to: Commanding Officer, Picatinny Arsenal, Dover, New Jersey, ATTN: SMUPA-NI.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specification, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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Preparing activity:
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